

Open Problems for Middle School

TCE 412M · Guest session with Todd Edwards

Unsolved problems your students can understand, try, and make real conjectures about

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THE IDEA

Collatz works in a middle-school classroom because it is a real open problem that anyone can start on: pick a number, follow a rule, watch what happens. Each problem below has that same shape, and each sits on content you will teach: primes, factors, fractions, rates. Nobody knows the answers, so your students are doing genuine mathematics. They gather evidence, spot patterns, and argue for conjectures of their own, instead of rediscovering a textbook result.

WHAT MAKES AN OPEN PROBLEM CLASSROOM-READY

1. **Easy to state.** A 12-year-old can understand the claim.
2. **Easy to try by hand.** Small cases take a pencil, not a proof.
3. **Better with technology.** A spreadsheet or applet checks thousands of cases and reveals patterns no one would see by hand.
4. **Something to look at.** The data makes a picture: a plot, a triangle, a spiral.

EIGHT PROBLEMS

Goldbach's Conjecture ↗

PRIMES · GR 6–8

Every even number greater than 2 is the sum of two primes.

Why it works: Checking one number is easy ($10 = 3 + 7 = 5 + 5$), but counting *how many ways* each number works, and plotting the counts, produces the striking “Goldbach comet”: the data fans out in bands nobody predicted.

With technology: Pick an even number, see every prime pair; slide up to 10,000 and watch the comet grow.

Open since 1742. Checked by computer up to 4×10^{18} .

Twin Prime Conjecture ↗

PRIMES · GR 6–8

There are infinitely many pairs of primes that differ by 2, like 11 and 13 or 41 and 43.

Why it works: Twins seem to thin out as numbers grow, yet they keep turning up. A running count lets students argue from evidence about whether they ever stop, which is the heart of the problem.

With technology: A number line or Ulam spiral that lights up twin pairs, with a live count and a chart of the gaps between primes.

Open. In 2013 Yitang Zhang proved some gap under 70 million repeats forever; that bound is now down to 246.

Legendre's Conjecture ↗

PRIMES · GR 7–9

There is always at least one prime between any two consecutive perfect squares.

Why it works: It connects primes to squares, which students can draw. The number of primes in each gap grows, but unevenly, so there is always a new question: which gap is the most crowded? The emptiest?

With technology: Slide n and see the interval from n^2 to $(n+1)^2$ drawn as a strip, with its primes marked and counted.

Open. With Goldbach and twin primes, one of Landau's four “unattackable” problems of 1912.

Gilbreath's Conjecture ↗

PRIMES · GR 5–8

List the primes, then take differences between neighbors again and again. Every row after the first starts with 1.

Why it works: The whole thing is subtraction, and it builds a triangle that is almost begging to be colored in. It is the most surprising problem here for the least mathematical background.

With technology: A growing difference triangle, colored by value, that students can extend row by row, or try with other starting lists to see if it still works.

Open since 1958. Checked for the first 3.4×10^{11} primes.

The 196 Problem ↗

ARITHMETIC · GR 4–7

Reverse a number's digits and add: $57 + 75 = 132$, $132 + 231 = 363$, a palindrome. Starting from 196, does it ever reach one?

Why it works: Only addition is needed, so it fits the youngest students here. Most numbers finish in a few steps, which makes 196 all the stranger. Like Collatz, each starting number has its own "journey."

With technology: Step through the reverse-and-add journey, then map which starting numbers finish fast, slowly, or seemingly never.

Open. Computers have followed 196 for hundreds of millions of digits without a palindrome.

Erdős–Straus Conjecture ↗

FRACTIONS · GR 5–8

For every whole number $n \geq 2$, the fraction $4/n$ can be written as $1/x + 1/y + 1/z$.

Why it works: It is a research problem about ordinary fractions and unit fractions, a weak spot for many students and teachers. Hunting for solutions is real fraction practice with a purpose.

With technology: Pick n , search for (or try to guess) the three unit fractions, and see them as pieces of a fraction bar.

Open since 1948. Checked for every n up to 10^{17} .

Odd Perfect Numbers ↗

FACTORS · GR 5–8

A number is perfect if its factors (not counting itself) add up to it: $6 = 1 + 2 + 3$. Every known perfect number is even. Is there an odd one?

Why it works: It rests on factors, a core middle-school topic. Along the way students sort numbers as "abundant" or "deficient" and find patterns in which ones are which.

With technology: Show each number's factors as bars that stack up against the number itself; color numbers by abundant, deficient, or perfect.

Open for over 2,000 years. No odd perfect number exists below 10^{1500} .

Lonely Runner Conjecture ↗

MOTION · GR 7–10

Runners circle a track, each at a different constant speed. Will every runner, at some moment, be far from all the others?

Why it works: It is the most visual problem on this list: an animation. Students can set speeds, run the race, and watch for the "lonely" moment, then connect what they see to fractions and rates.

With technology: An animated circular track with adjustable speeds that flags each runner's loneliest moment.

Open in general; proved when there are only a few runners (seven or fewer, as of 2008).

WHERE TO START WITH A CLASS

The 196 Problem

Only addition, so every student can play. Assign each group a range of starting numbers and pool the results.

Gilbreath's Conjecture

Just subtraction, and the triangle can be built on the board as a whole class. The surprise lands every time.

Goldbach's Conjecture

Great practice with primes. Have students find every pair for their own number, then graph the class's data.

Click any problem name for its page, picture, and a detailed one-pager: ohiomathteacher.github.io/applet-library/open-problems/

Whatever the problem: have students predict before they compute, keep a log of cases, and write the conjecture their evidence supports, along with what would change their mind.